

Supporting Information for: Plant-wide Bayesian fault diagnosis in recycle processes via simulation-based inference under multi-loop feedback control

Simone Casolo ^{1,*} Alexander Johannes Stasik ^{2,3,†} and Signe Riemer-Sørensen ³

¹*Cognite AS, Oslo, Norway*

²*Department of Data Science, Norwegian University of Life Sciences, Ås, Norway*

³*Department for Mathematics and Cybernetics, SINTEF Digital, Oslo, Norway*

(Dated: July 15, 2026)

S1. STOCHASTIC DATA GENERATION AND SCENARIO OVERVIEW

a. Simulation pipeline. Observation windows are generated by integrating the degraded closed-loop model (Eqs. (3)–(6) of the main text) with the Euler–Maruyama scheme at a fixed step of $\Delta t = 0.1$ min. Two independent noise layers are applied. *Process noise* is injected at each integration step as an additive Gaussian increment: $\sigma = [\sigma_C, \sigma_T, \sigma_{T_c}] = [5 \times 10^{-4} \text{ mol L}^{-1} \text{ min}^{-1/2}, 0.1 \text{ K min}^{-1/2}, 0.1 \text{ K min}^{-1/2}]$, consistent with mild process variability at steady state. *Sensor noise* is applied post-simulation as a zero-mean Gaussian draw with standard deviation equal to 0.5% of each channel’s running maximum, representing a typical industrial-grade sensor specification.

Each replicate is initialised from the nominal closed-loop steady state, computed by integrating the deterministic ODE to convergence with a high-accuracy adaptive solver. Initialising from the steady state removes startup transients and ensures that the 60-minute window is representative of stationary operation around the prescribed (α^*, β^*) operating point. The four observable channels $[C(t), T(t), T_c(t), Q_c(t)]$ are subsampled to a grid of $\Delta t_{\text{out}} = 0.5$ min, yielding one 120×4 observation matrix per replicate.

b. Dataset structure. The full System I dataset comprises 8 scenarios $\times$ 50 independent stochastic replicates = 400 observation windows. The eight scenarios (Table 1 of the main text) span open- and closed-loop configurations and a range of fault severities; their ground-truth parameters (α^*, β^*) are summarised in Table 1 of the main text. Each replicate within a scenario shares the same ground-truth (α^*, β^*) but was integrated with an independent random seed, so the 50 replicates per scenario are exchangeable draws from $p(\mathbf{x} \mid \alpha^*, \beta^*)$ at that operating point.

c. Scenario trajectories. Figure 1 shows six representative replicates per closed-loop scenario across all four observable channels. The bold black trace is the per-scenario replicate mean.

Several features are immediately apparent. First, the reactor temperature T is tightly regulated near the set-point $T_{\text{sp}} = 312.5$ K in all non-saturating scenarios: the

PI controller maintains the same temperature profile regardless of whether the jacket is fouled (Sc 2) or the catalyst is decaying (Sc 3), confirming the closed-loop masking mechanism discussed in Section 2.2 of the main text. Second, the coolant flow Q_c shows qualitatively distinct behaviour across scenarios: jacket fouling (Sc 2) drives Q_c to $\approx 210 \text{ L min}^{-1}$, while catalyst decay (Sc 3) lowers it to $\approx 50 \text{ L min}^{-1}$, and severe fouling (Sc 5) saturates the valve at $Q_{c,\text{max}} = 400 \text{ L min}^{-1}$. Third, the concentration channel C shifts distinctly between scenarios with different α^* : catalyst decay (Sc 3, $\alpha^* = 0.70$) raises C_{ss} from ≈ 0.018 to $\approx 0.026 \text{ mol L}^{-1}$, while jacket fouling (Sc 2) produces a modest downward shift attributable to the residual thermal coupling through T_c . The sensor-drift scenario (Sc 7, $\delta_T = +2$ K) is indistinguishable from Sc 1 in the C and T_c channels but shows a systematic upward offset in the T channel, motivating the four-dimensional $(\alpha, \beta, \delta_T, \delta_{C_i})$ extension explored in future work.

S2. PRIOR PREDICTIVE CHECK

A necessary condition for valid amortised inference is that the prior-predictive distribution — the distribution of summary statistics produced by prior draws — covers the evaluation observations. To verify this, we drew 500 parameter vectors $\theta = (\alpha, \beta)$ from the uniform prior $\mathcal{U}([0.4, 1.2]^2)$ and simulated each through the stochastic closed-loop model to obtain synthetic 60-minute observation windows. For every feature in the 29-D summary vector, the [2.5%, 97.5%] interval of the prior-predictive distribution was compared against the observed values from all six closed-loop evaluation scenarios (Sc 1–5, Sc 7). At least 80% of the observed values per feature fall inside this interval, confirming that the simulator faithfully spans the evaluation regime.

Figure 2 illustrates the check for four representative features, including the two physics-informed proxies that carry most of the identifying information about (α, β) (see Section 3.1 of the main text). The observed scenario means (vertical markers) are well contained within the prior-predictive histograms for every feature shown, with no scenario lying systematically outside the prior envelope.

* simone.casolo@cognite.com

† alexander.stasik@sintef.no

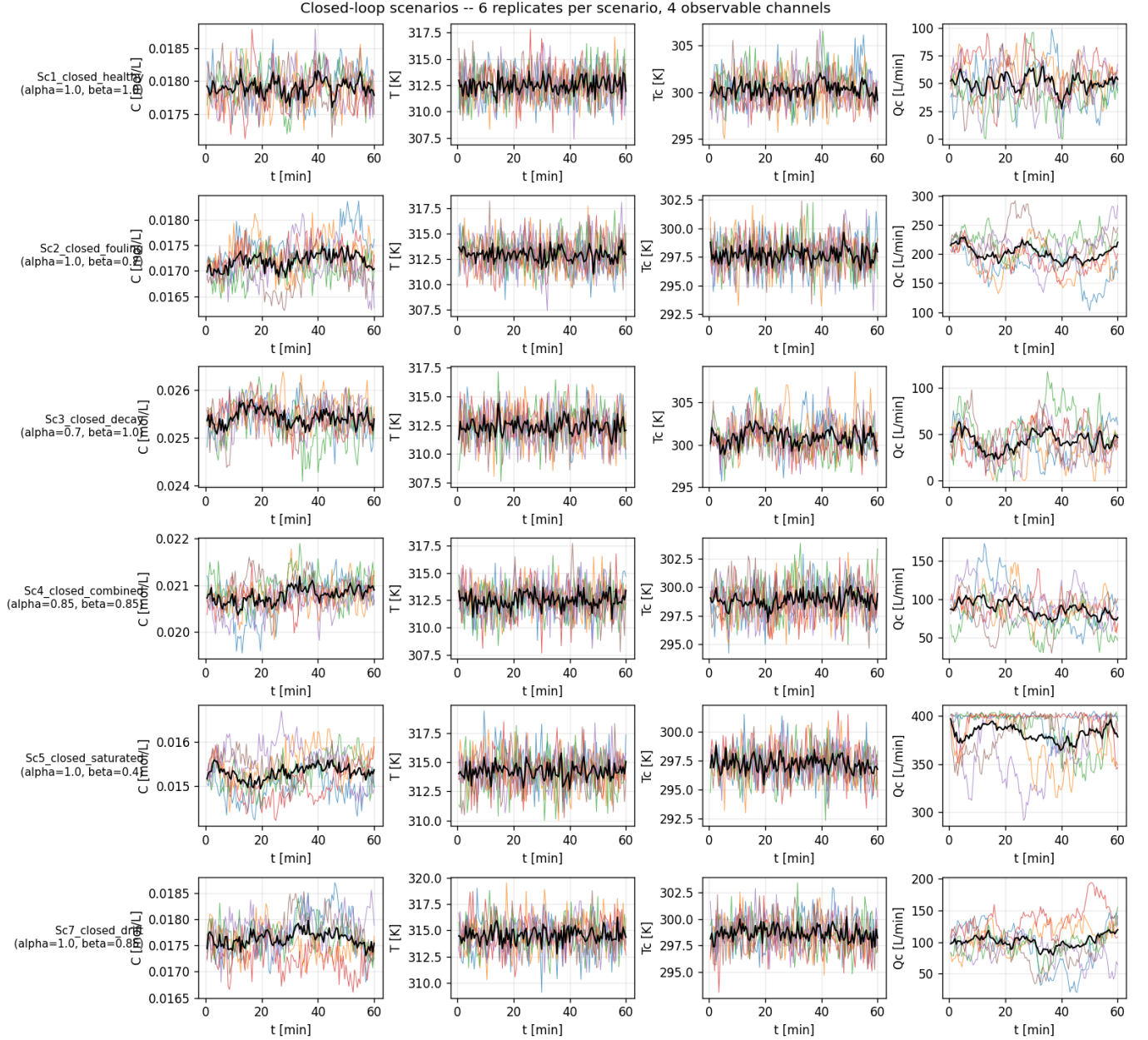


FIG. 1: Representative observation windows for the six closed-loop fault scenarios (Sc 1–5, Sc 7). Each row corresponds to one scenario (scenario name and ground-truth (α^*, β^*) shown on the left); columns show the four observable channels $[C, T, T_c, Q_c]$ over the 60-minute window. Coloured traces: six individual stochastic replicates. Bold black trace: replicate mean. The reactor temperature T (second column) is regulated near $T_{sp} = 312.5$ K across all scenarios; the fault signal is encoded primarily in Q_c (fourth column) and C (first column), consistent with the Fisher-information analysis in Section 4.3.1 of the main text.

S3. TRAINING BUDGET SENSITIVITY

To confirm that the choice of $n_{sim} = 10,000$ training simulations is sufficient, we trained three posteriors using MAF density estimators at budgets of 1,000, 5,000, and 10,000 prior draws and evaluated each on the healthy scenario (Sc 1, $\alpha^* = \beta^* = 1.0$) and the jacket-fouling scenario (Sc 2, $\beta^* = 0.70$).

All three budgets achieve 90% credible-interval coverage of the true parameters on Sc 1. For Sc 2, the posterior mean of β shifts monotonically from $\hat{\beta} = 0.680$ at $n_{sim} = 1,000$ to $\hat{\beta} = 0.618$ at $n_{sim} = 10,000$, while 90% CI coverage is attained at all three budgets. Increasing the budget beyond 10,000 produced no measurable improvement in the simulation-based calibration metrics (Section S6), motivating the choice of $n_{sim} = 10,000$ for

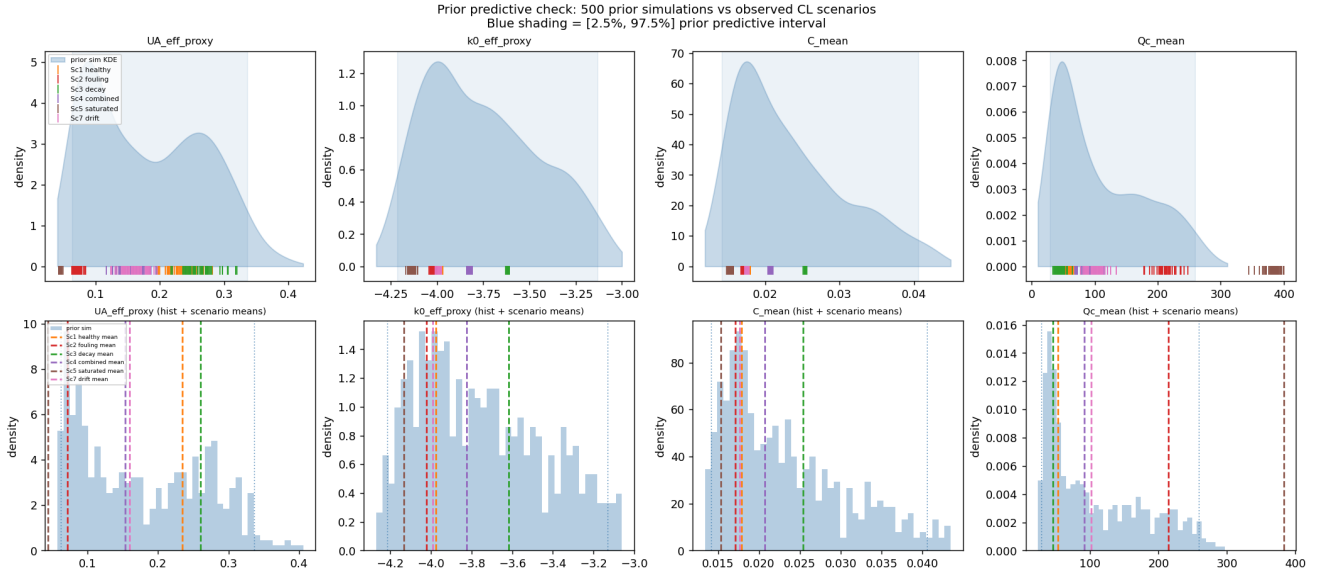


FIG. 2: Prior predictive check ($N = 500$ prior draws) for four key summary features: s_{UA} , s_{k_0} , $\bar{C}$, and $\bar{Q}_c$. *Top row*: kernel density estimates of the prior-predictive distribution (blue fill) with observed values from each closed-loop scenario overlaid as tick marks. *Bottom row*: histogram of prior-predictive values with per-scenario means marked as vertical lines. All scenario observations fall within the prior-predictive envelope, confirming that the simulator covers the operating regime of the evaluation set.

the production NSF posterior.

The persistent $\hat{\beta}$ bias observed across all three budgets is not a training artefact: it is the structural closed-loop masking effect characterised analytically in Section 4.3 of the main text and confirmed by four independent inference methods.

S4. PCA MANIFOLD OF SUMMARY STATISTICS

To complement the physics-proxy scatter plot shown in the main text, Figure 4 shows the principal component analysis (PCA) projection of the standardised 29-D summary vectors across all 400 observation windows (8 scenarios, 50 replicates each).

All eight scenarios occupy visually distinct regions in the PC1–PC2 projection, which together account for 57.7% of the total variance. The severe-fouling scenario (Sc5) separates primarily along PC1, driven by the saturation-fraction features that are identically zero in all other scenarios. The open-loop scenarios (Sc0, Sc6; squares) are resolved from their closed-loop counterparts in the PC3–PC4 projection, where the standard deviation and slope of Q_c provide the distinguishing signal.

This global separability confirms that the 29-D vector encodes adequate information for all eight scenarios, including regimes — such as valve saturation and open-loop operation — where the two physics-proxy features s_{UA} and s_{k_0} lose discriminating power.

S5. COMPLETE SUMMARY-STATISTIC DEFINITIONS AND FEATURE ABLATION

Table I lists all 29 features that form the summary-statistic vector $\mathbf{s} \in \mathbb{R}^{29}$ used throughout this work. The four groups correspond to those described in Section 3.1 of the main text.

Table II reports five-fold stratified cross-validated LDA classification accuracy for seven feature subsets on the 400-window System I dataset. The physics-2 subset (Group 4, two features) alone achieves 98.3% accuracy, confirming that the steady-state proxy features encode nearly all scenario-separating information. The remaining 27 features are necessary not to improve classification accuracy — which saturates at 100% with as few as four final-window means — but to correctly quantify posterior uncertainty under noise, transients, and actuator saturation. This distinction between classification accuracy and posterior calibration is central to the argument of the main text: the 29-D vector is chosen to provide accurate full posterior distributions, not merely correct fault labels.

Beyond the two-feature theoretical minimum, four practical effects require additional features: sensor noise amplifies in the ratio s_{UA} at moderate Q_c values, making it preferable to supply the three component channels separately; slope and final-window features carry transient information during fault onset; the saturation-fraction features take over from s_{UA} when the valve is fully open (Sc5); and the standard deviation of Q_c distinguishes open-loop windows (constant Q_c) from closed-loop ones.

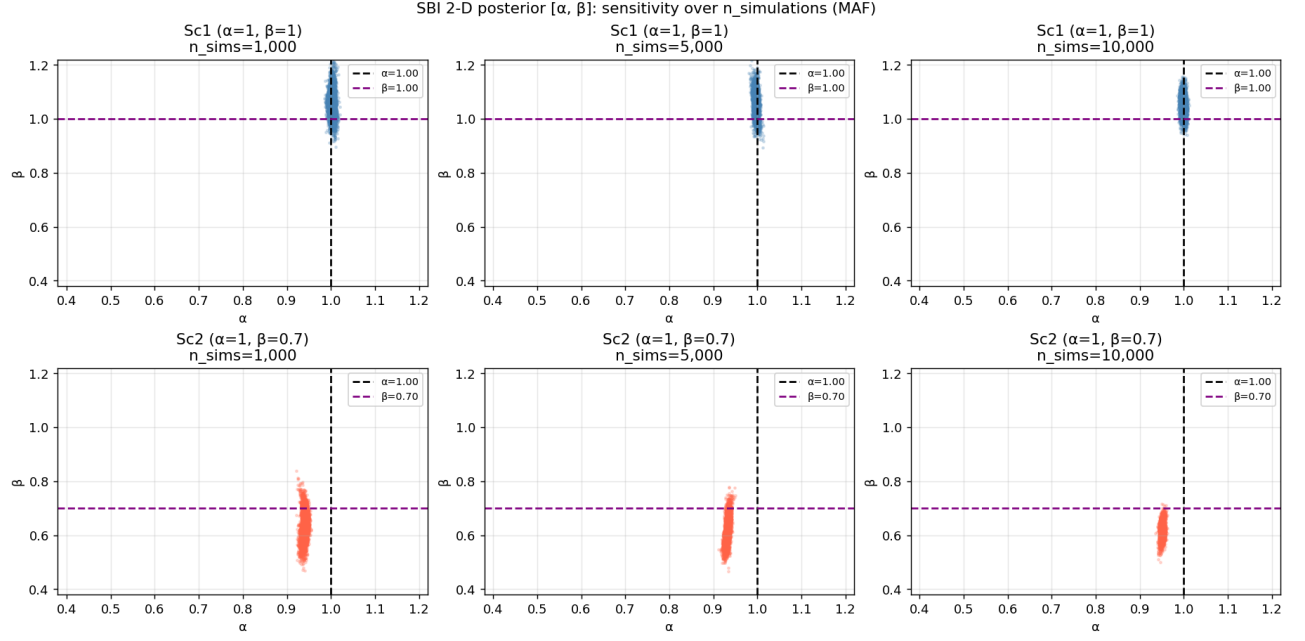


FIG. 3: Training budget sensitivity: 2-D posterior scatter plots $[\alpha, \beta]$ for Sc 1 (healthy, top row) and Sc 2 (jacket fouling, bottom row) at three simulation budgets (MAF estimator). Dashed lines mark the true parameter values. All three budgets achieve 90% credible-interval coverage; the posterior concentrates monotonically with n_{sim} , with no change in the systematic β underestimation that characterises all inference methods on this closed-loop dataset.

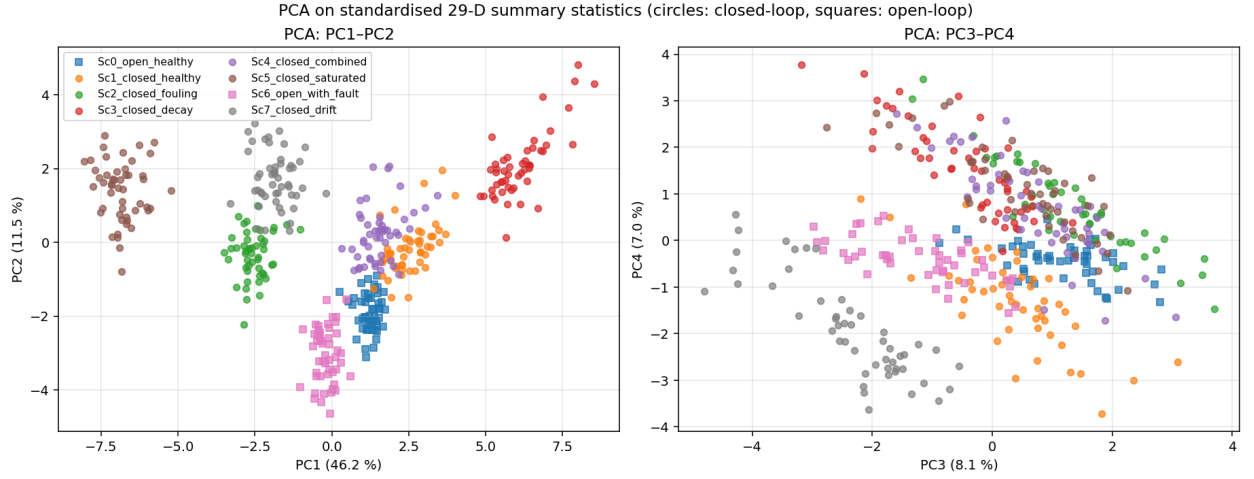


FIG. 4: PCA of the standardised 29-D summary-statistic vectors (400 windows, 8 scenarios). Circles: closed-loop replicates; squares: open-loop replicates (Sc0, Sc6). *Left*: PC1–PC2 projection (46.2% + 11.5% variance explained). *Right*: PC3–PC4 projection further resolves the open-loop scenarios from their closed-loop counterparts.

TABLE I: Complete summary-statistic vector for System I: 29 features in four groups. $(\bar{\cdot})$ = whole-window mean; $(\cdot)_f$ = final-25% window mean; channel indices $[C, T, T_c, Q_c]$.

Group	Features	Count	Physical motivation
1	mean, std, slope, min, max $\times [C, T, T_c, Q_c]$	20	Level, variability, trend
2	$C_f, T_f, T_{c,f}, Q_{c,f}$	4	Quasi-steady-state estimate
3	$\int T - T_{sp} dt, f_{\text{sat, low}}, f_{\text{sat, high}}$	3	Controller load / saturation
4	s_{UA}, s_{k0}	2	Physics-informed proxies
Total		29	

TABLE II: Feature-subset ablation: five-fold stratified cross-validated LDA accuracy on the 400-window System I dataset (8 scenarios, 50 replicates each). MI = mutual information with scenario label. The saturation of accuracy at 100% for several subsets shows that classification difficulty does not determine the choice of feature set; posterior calibration does.

Feature subset	n features	CV accuracy
Full 29-D vector	29	1.000
Top-10 MI shortlist	10	0.991
Top-5 MI shortlist	5	0.979
Group 4 (physics-2)	2	0.983
Group 3 (control aggregates)	3	0.800
Group 2 (final-window means)	4	1.000
Group 1 (per-channel base)	20	0.998

S6. POSTERIOR CALIBRATION: SIMULATION-BASED CALIBRATION PROTOCOL AND RESULTS

d. Protocol. Posterior calibration is assessed using simulation-based calibration [?]. In SBC, N_{SBC} parameter vectors $\theta^{(i)}$ are drawn independently from the prior; for each draw a synthetic observation $\mathbf{x}^{(i)}$ is simulated and the posterior is sampled to obtain L approximate draws $\{\tilde{\theta}_\ell^{(i)}\}_{\ell=1}^L$. The rank of the true parameter among the posterior draws,

$$r_j^{(i)} = \sum_{\ell=1}^L \mathbf{1}\{\tilde{\theta}_{\ell,j}^{(i)} < \theta_j^{(i)}\}, \quad (1)$$

is recorded for each marginal $j \in \{\alpha, \beta\}$. A well-calibrated posterior yields rank statistics that are uniformly distributed on $\{0, 1, \dots, L\}$ across all N_{SBC} test cases. Uniformity is assessed via the Kolmogorov–Smirnov (KS) test and the two-sample classifier score (C2ST; ?): a KS p -value above 0.05 indicates that uniformity cannot be rejected at the 5% level; a C2ST score below 0.52 indicates that the rank distribution is indistinguishable from uniform by a trained binary classifier [1]. We use $N_{\text{SBC}} = 500$ test cases and $L = 1,000$ posterior draws per case.

e. Results. The KS p -values for the production NSF posterior are 0.016 (α) and 0.014 (β), formally rejecting rank uniformity at the 5% level for both marginals (Figure 5). The C2ST scores are 0.52 (α) and 0.53 (β), both near the uninformative baseline of 0.5.

f. Interpretation. The combination of $p < 0.05$ on the KS test with C2ST ≈ 0.5 is the diagnostic signature of a structural information deficit rather than a failure of the density estimator. The KS test rejects uniformity because the rank distribution is systematically compressed toward low values for β : the PI controller enforces $T = T_{\text{sp}}$, removing the temperature channel’s contribution to $I_{\beta\beta}$ and producing a posterior that concentrates below the true β . The C2ST score remaining

near 0.5 confirms that this compression is subtle and the estimator is performing correctly given the information in the data. A miscalibration arising from a training deficiency — insufficient simulations, architecture mismatch, or a distributional gap between training and evaluation — would instead produce both a low KS p -value *and* a C2ST score substantially above 0.5. The fact that neither C2ST score exceeds 0.53 therefore supports the interpretation that the $n_{\text{sim}} = 10,000$ production posterior is well-trained and that the miscalibration is irreducible from closed-loop data, consistent with the Cramér-Rao argument and the embedding-net irreducibility experiment in Section 4.3.3 of the main text.

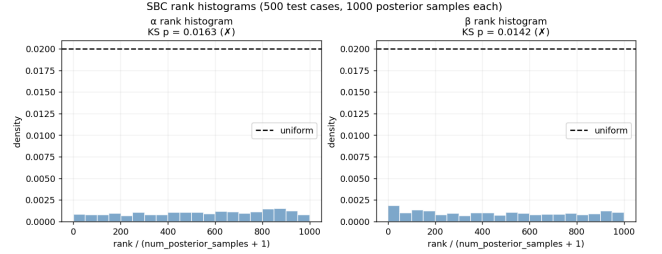


FIG. 5: SBC rank histograms for the production NSF posterior ($N_{\text{SBC}} = 500$ test cases, $L = 1,000$ posterior draws each). Dashed line: expected uniform density. Both marginals are compressed toward low ranks (KS $p = 0.016$ for α , $p = 0.014$ for β), reflecting the systematic downward shift of the β posterior caused by the PI controller’s suppression of the temperature information channel. The C2ST scores of 0.52 and 0.53 confirm the miscalibration is subtle and consistent with a structural Fisher-information deficit, not a training deficiency.

S7. SYSTEM II: FAULT SCENARIOS AND DATA-GENERATION DETAILS

S7.1 Nominal parameter values

S7.2 Fault scenario catalogue

S7.3 Data generation and training

Observation windows are generated using the same Euler–Maruyama protocol as System I (Section S1), adapted to the Wu 2003 plant dynamics. The integration step is $\Delta t = 0.01$ h, with observations subsampled to 1-min resolution ($\Delta t_{\text{out}} = 1/60$ h), giving 120 time steps per 2-hour window. Process noise diffusion coefficients and sensor noise level (0.3% of channel range for the recycle plant, reflecting tighter instrumentation relative to the laboratory-scale System I) were chosen to produce a steady-state signal-to-noise ratio comparable to System I. Each replicate is initialised from the nomi-

nal closed-loop steady state computed by integrating the deterministic ODE to convergence.

The production posterior for Structure S-B was trained using the `zuko` normalising-flow backend (`zuko_nsf`, accessed via the `sbi` library [1?]) with a neural spline flow architecture (60 hidden units, 3 transforms, $n_{\text{sim}} = 15,000$ prior draws), trained with the Adam optimiser at the `sbi` library’s default batch size of 200 for up to 200 epochs with early stopping (patience 20 epochs) on a held-out 10% validation split — the same stopping protocol as System I (Section 4, main text), but a smaller network and a batch size that was not explicitly re-tuned from the library default. This architecture was chosen from a sweep including larger (80 hidden units/5 transforms, 128/5) and smaller (30/2, 40/2) configurations: larger networks produced markedly worse calibration at this parameter dimensionality (in some cases uniform miscalibration across all five parameters), consistent with over-parameterisation relative to the 15,000-sample training bank, ruling out “the network is too small” as an explanation for any residual calibration difficulty. Training is additionally seed-unstable at this dimensionality: identical training data and architecture, differing only in the random initialisation, can flip a parameter’s simulation-based calibration outcome from passing to failing. An 8-seed ensemble with multi- N simulation-based calibration (Section S6) was therefore used to select the production posterior (Seed 4), rather than training a single posterior as for System I. Structure S-A failed calibration across 40 seeds and architectures (including the same larger/smaller sweep described above, and principal-component feature-dimension reduction to 15 or 25 components) and is reported only as a negative result (Section 7 of the main text).

S8. MULTI-METHOD BASELINE: IMPLEMENTATION DETAILS AND PER-REPLICATE DISTRIBUTIONS

Table 1 of the main text (Section 4.4) summarises the four-method comparison on Sc2. This section provides the implementation details and per-replicate posterior distributions that underpin that table.

The No-U-Turn Sampler (NUTS) Markov chain Monte Carlo [?] was implemented in NumPyro [?] with 200 warmup and 300 draw steps across two chains, using the full-rank dense mass matrix and a `difffrax` Tsit5 adaptive ODE solver for the trajectory likelihood; convergence was confirmed by $\hat{R} \approx 1.00$ and effective sample size > 400 for all parameters. The augmented-state extended Kalman filter (EKF) was implemented with an analytically derived 6×6 Jacobian (augmented state: $[C, T, T_c, I, \alpha, \beta]$) and the unscented Kalman filter (UKF) with 13 sigma points. Noise covariances for both filters were tuned on 50 healthy (Sc1) replicates.

Figure 7 shows the per-replicate $\hat{\beta}$ distributions for all four methods alongside the inference timing comparison.

All four distributions are centred below the true value $\beta^* = 0.70$ by a comparable offset, confirming that the structural bias is independent of the inference paradigm.

S9. SYSTEM II: SUMMARY-STATISTIC FEATURE VECTOR AND CALIBRATION

S9.1 Feature vector definition

The 66-D summary-statistic vector for Structure S-B is constructed from the nine observable channels $[T_r, T_j, Q_j, T_{\text{reb}}, Q_{\text{reb}}, F_{R,\text{norm}}, F_{B,\text{norm}}, R_{\text{norm}}, V_{\text{norm}}]$ using the same four-group framework as System I. Structure S-A adds x_D as a tenth channel (72-D total).

Group 1 — Per-channel base statistics (45 features for S-B, 50 for S-A). For each observable channel, five scalar statistics are computed over the full 2-hour window: mean, standard deviation, ordinary-least-squares slope, minimum, and maximum.

Group 2 — Final-window means (9 or 10 features). Arithmetic mean of each channel over the final 25% of the 2-hour window, providing a quasi-steady-state estimate less contaminated by onset transients.

Group 3 — Control aggregates (6 features). Time-integrated absolute temperature errors for Loop 1 and Loop 3 ($\int |T_r - T_{\text{sp}}| dt$, $\int |T_{\text{reb}} - T_{\text{reb,sp}}| dt$), valve-saturation fractions for both manipulated variables, and running standard deviations of $F_{R,\text{norm}}$ and $F_{B,\text{norm}}$ (snowball-dynamics indicators).

Group 4 — Physics-informed proxies (6 features). Six analytically motivated features encoding the identifiable parameter combinations: $s_{U_{A,r}} = (T_r - T_j)/Q_j$ (Eq. (18) of the main text); $s_{F_R} = F_{R,\text{norm}} - 1$ (snowball indicator, Eq. (19)); per-pass conversion proxy $s_{\text{conv}} = \ln(F_R/(F_0 z_0))$; reboiler heat proxy $s_{\text{reb}} = Q_{\text{reb}}/V_{\text{norm}}$; and two cross-channel correlation features $\text{corr}(Q_j, F_{R,\text{norm}})$ and $\text{corr}(Q_{\text{reb}}, F_{R,\text{norm}})$, which capture the co-variation between reactor cooling demand and recycle load that discriminates α -related faults from β_r -related ones.

S9.2 Posterior calibration

As described in Section S7.3, an 8-seed ensemble was trained for Structure S-B; Seed 4 passed simulation-based calibration at $N_{\text{SBC}} \in \{200, 400, 800\}$ for all five parameters. Structure S-A failed calibration in all 40 seeds tested and is treated as a settled negative result. The S-B rank histograms and KS p -values are shown in Figure 8.

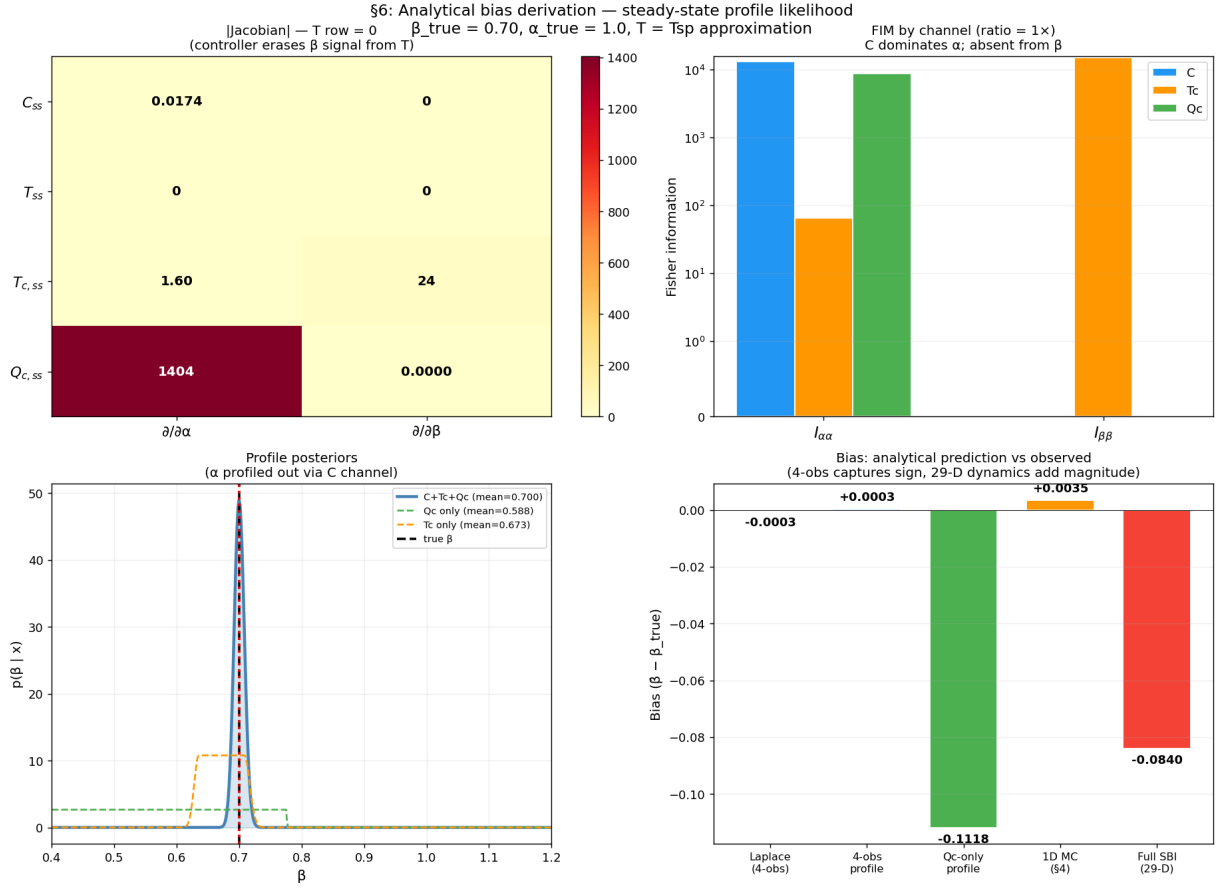


FIG. 6: Analytical characterisation of the closed-loop β bias. *Top left*: absolute Jacobian $|\partial \bar{s}_i / \partial \theta_j|$ for the four steady-state observables. The temperature row is identically zero for both parameters; the concentration column is zero for β (structural decoupling). *Top right*: Fisher information decomposition by channel. $I_{\alpha\alpha}$ is dominated by the concentration channel (>99%); $I_{\beta\beta}$ relies entirely on the noisier T_c and Q_c channels, yielding a ratio of $\approx 600\times$ from the analytical 4-observable model. *Bottom left*: Profile posterior $p(\beta | \mathbf{x})$ from the analytical steady-state model (all four channels, Q_c only, T_c only). The asymmetric likelihood produces a small negative bias even in this simplified model. *Bottom right*: Comparison of predicted and observed bias magnitudes across five analysis variants; the full 29-D dynamic summary space amplifies the analytically predicted skewness.

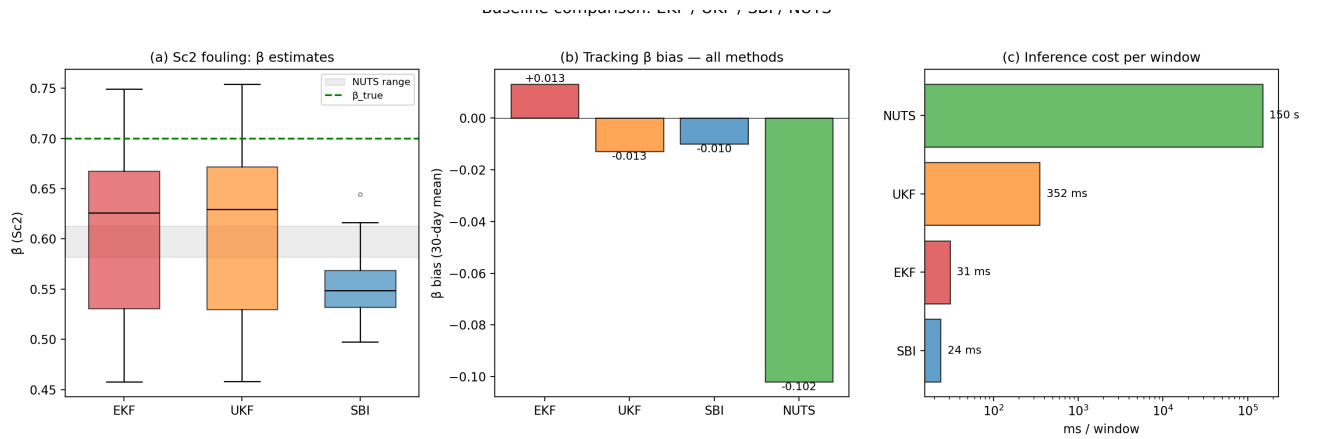


FIG. 7: Multi-method baseline comparison (System I, Sc 2, $\alpha^* = 1.0$, $\beta^* = 0.70$, 50 replicates). *Left panels*: $\hat{\beta}$ distributions for all four methods; dashed line marks $\beta^* = 0.70$. *Top right*: mean β bias per method. *Bottom right*: per-window inference time (log scale), illustrating the $> 9,000\times$ speed advantage of amortised SBI over NUTS.

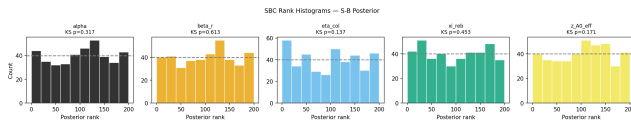


FIG. 8: Simulation-based calibration rank histograms for the calibrated System II S-B posterior (Seed 4, $N_{\text{SBC}} = 800, 1,000$ posterior draws per case) for all five parameters. All marginals pass the KS uniformity test at the 5% level, confirming the posterior is well-calibrated for α , β_r , η_{col} , ξ_{reb} , and $z_{A0,\text{eff}}$.

S10. SYSTEM II: CNN EMBEDDING STUDY — FULL RESULTS

Section 5.2 of the main text summarises the CNN embedding experiment (nb32) in two sentences. This section provides the full results.

S10.1 Architecture and calibration

A convolutional neural network (CNN) embedding was trained on raw 120×9 S-B time-series tensors and connected to the same normalising-flow density estimator used for the hand-crafted posterior (NSF, 60 hidden units, 3 transforms, $n_{\text{sim}} = 15,000$). The CNN maps each 120×9 window to a 30-dimensional latent vector, replacing the hand-crafted 66-D summary statistics entirely. An 8-seed ensemble was trained and the best seed was selected by the same multi- N SBC protocol as the production hand-crafted posterior (Section S7.3). The selected CNN posterior passes SBC calibration at $N_{\text{SBC}} \in \{200, 400, 800\}$ for all five parameters.

S10.2 Artifact-scenario correlations

Table III compares the joint-posterior correlations at the three artifact scenarios between the hand-crafted and CNN posteriors, pooled across all 30 replicates at each scenario. The CNN does not resolve the (α, β_r) coupling at W11 (correlation +0.993 vs. hand-crafted +0.998); a follow-up grid-check (15 noise seeds at W11 plus 4 additional (α, β_r) truth points) confirms this null result is not a single-point artefact, with correlation remaining in $[0.987, 0.996]$ across all 19 tested conditions. Artifact 1 $(\alpha, \eta_{\text{col}})$ at W12 remains near-zero under both representations, consistent with the earlier retraction. Artifact 3 $(\alpha, z_{A0,\text{eff}})$ at W10 collapses under the CNN, from +0.995 (hand-crafted) to -0.004 (CNN) — despite this, feed classification does not improve (§10.3), because the CNN posterior introduces its own systematic bias that produces a different misclassification rather than a correct one.

TABLE III: Joint-posterior correlations at the three artifact scenarios: hand-crafted 66-D vs. CNN 30-D. Correlations pooled across all 30 replicates.

Scenario	Parameter pair	Hand-crafted	CNN
W11 ($\alpha=0.80, \beta_r=0.80$)	(α, β_r)	+0.998	+0.993
W12 ($\alpha=0.75, \eta_{\text{col}}=0.80$)	$(\alpha, \eta_{\text{col}})$	+0.011	+0.010
W10 ($z_{A0,\text{eff}}=0.75$)	$(\alpha, z_{A0,\text{eff}})$	+0.995	-0.004
W13 ($\alpha=0.80, z_{A0,\text{eff}}=0.80$)	$(\alpha, z_{A0,\text{eff}})$	+0.993	-0.003

S10.3 Fault classification comparison

Table IV compares unit-level macro-F1 and per-class F1 between the two posteriors. Overall CNN performance is substantially worse than the hand-crafted posterior (macro-F1 0.270 vs. 0.694; accuracy 61.9% vs. 87.4%), not comparable. Healthy and column classification degrade the most: the CNN misclassifies all W1 (healthy) replicates as reactor, and all W7/W9 (column) replicates as multi. Feed-fault F1 remains 0.000 under both representations despite the CNN resolving the underlying $(\alpha, z_{A0,\text{eff}})$ correlation (§10.2) — the CNN posterior’s own systematic downward bias in α (present across nearly all scenarios, including healthy) independently pushes W10 samples past the reactor threshold, so the correlation’s resolution does not translate into correct classification.

TABLE IV: Unit-level fault classification: hand-crafted 66-D vs. CNN 30-D (14 scenarios, 30 replicates each).

Metric	Hand-crafted	CNN
Accuracy	87.4%	61.9%
Macro-F1	0.694	0.270
Healthy F1	0.667	0.000
Reactor F1	0.948	0.800
Column F1	1.000	0.000
Feed F1	0.000	0.000
Multi F1	0.854	0.548

S10.4 30-day progressive tracking

Figure 9 compares sequential degradation tracking over 30 days (360 windows at 2-hour cadence). The result is mixed rather than uniformly comparable: the CNN posterior improves β_r tracking relative to the hand-crafted posterior (MAE 0.198 $\rightarrow$ 0.139) but worsens α tracking (MAE 0.048 $\rightarrow$ 0.067). Both remain more than an order of magnitude worse than the EKF with raw-trajectory access (Section 7.5 of the main text, nb27; MAE 0.002 for α , 0.004 for β_r), as expected from a recursive estimator with privileged, exact-model access to the raw trajectory at every step, rather than an amortised posterior trained on a compressed or learned representation.

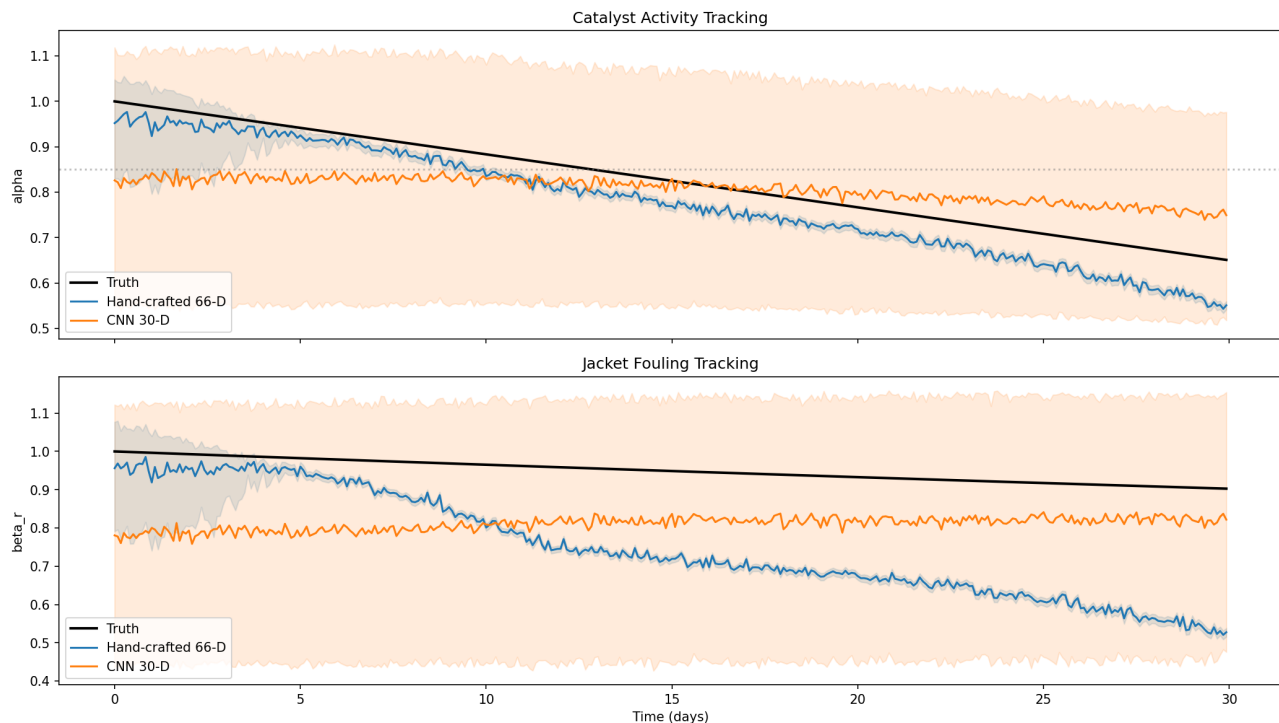


FIG. 9: 30-day progressive tracking of $\alpha(t)$ and $\beta_r(t)$: hand-crafted 66-D posterior (blue) vs. CNN 30-D posterior (orange), with 90% CI shading. True degradation profiles in black. Both methods track the gradual catalyst decay well; CNN provides similar MAE but wider credible intervals for β_r near the fouling saturation region.

S10.5 Interpretation

The CNN experiment establishes an important distinction: the information necessary to resolve the (α, β_r) degeneracy is present in the raw trajectory (demonstrated by the FIM analysis in Section 5.3 and the EKF experiment in Section 7.5), but a neural estimator trained at 15,000 draws on 120×9 inputs does not learn to extract it. This is not contradicted by the FIM result; rather,

it separates two different questions: *is the information physically present in the signal?* (yes, FIM and EKF confirm this) and *can a trained amortised posterior at this budget recover it?* (no, at least not at $n_{\text{sim}} = 15,000$ with the architecture tested). Improving the CNN posterior to close this gap — likely requiring substantially more training data, a deeper architecture, or an explicit inductive bias for the transient dynamics — is left to future work.

-
- [1] Á. Tejero-Cantero, J. Boelts, M. Deistler, J.-M. Lueckmann, C. Durkan, P. J. Gonçalves, D. S. Ziegler, K. Cranmer, and J. H. Macke, *sbi*: A toolkit for simulation-based inference, *Journal of Open Source Software* **5**, 2505 (2020).
 - [2] K.-L. Wu, C.-C. Yu, W. L. Luyben, and S. Skogestad, Reactor/separators processes with recycle — 2. design for composition control, *Computers & Chemical Engineering* **27**, 401 (2003).
 - [3] K. Cranmer, J. Brehmer, and G. Louppe, The frontier of simulation-based inference, *Proceedings of the National Academy of Sciences* **117**, 30055 (2020).